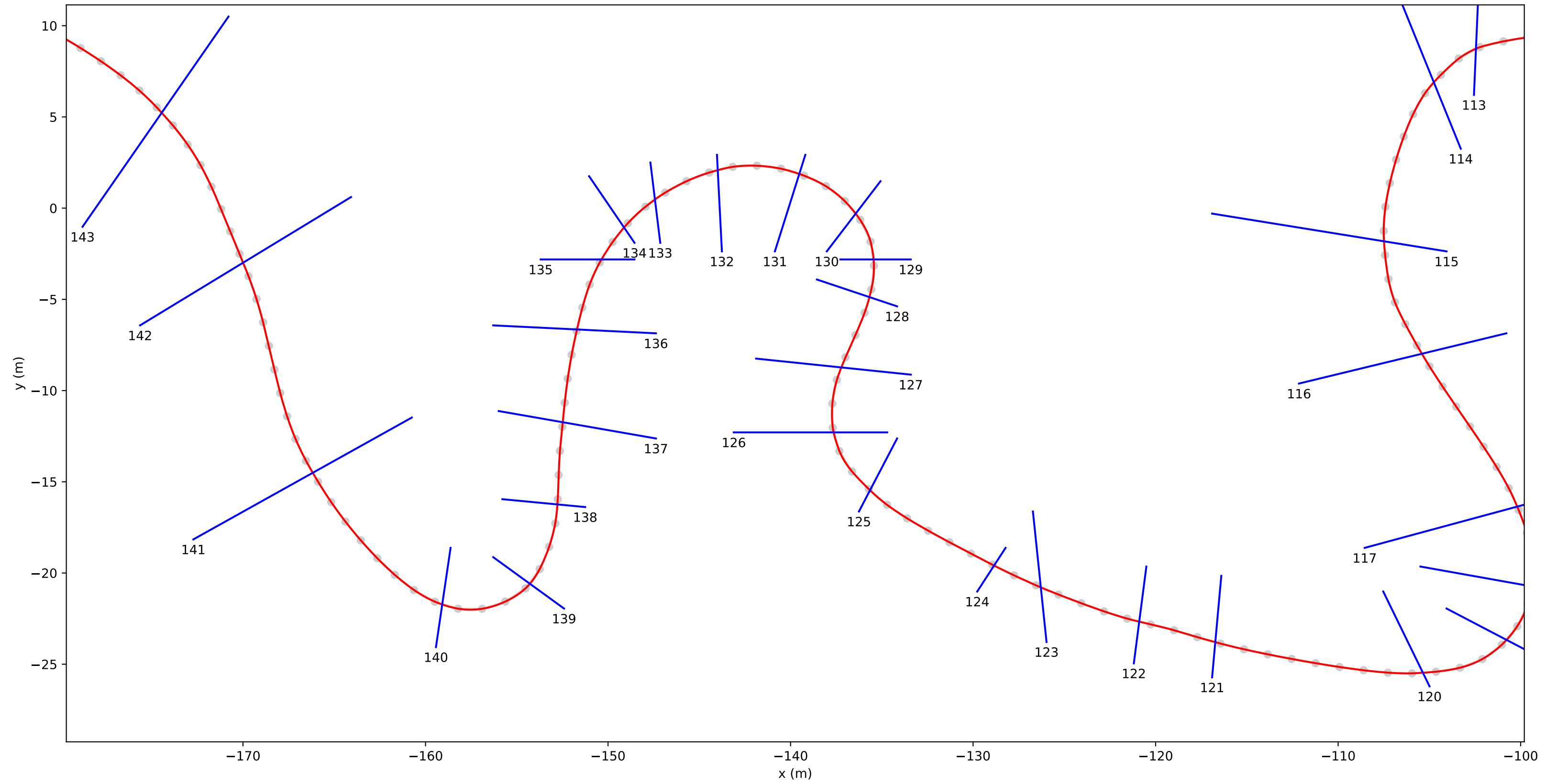


Map plot: optimized path through gates



Track Generation Report

=====

Generated: 2026-08-22 23:43:01

Gate file: ref_gates_enduranceTwist.csv

Gate count: 67

Saved DOFs file: opt_dofs_2026-08-22_23-43-01.npy

Saved DOF history file: dof_history_2026-08-22_23-43-01.npy

Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

Point counts

path generation points: 402

Results

total optimization time: 2678.850 s

total objective function evaluations: 41027

optimizer iterations (accepted steps): 35

total path length: 532.2854 m

total objective function value: 2.700204

Per-step time estimate (one measured objective evaluation)

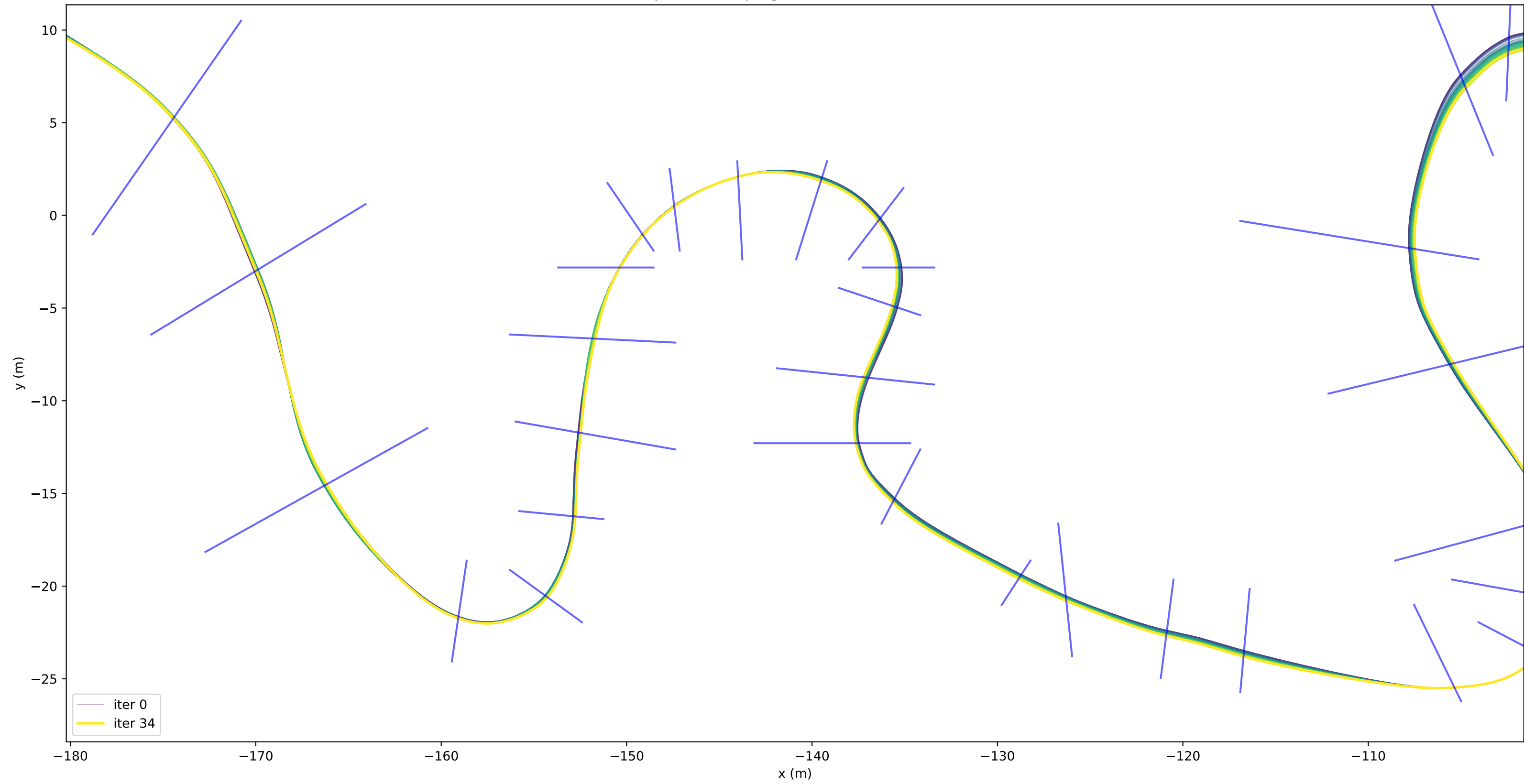
make spline from dofs: 6.13 ms

path length function: 0.36 ms

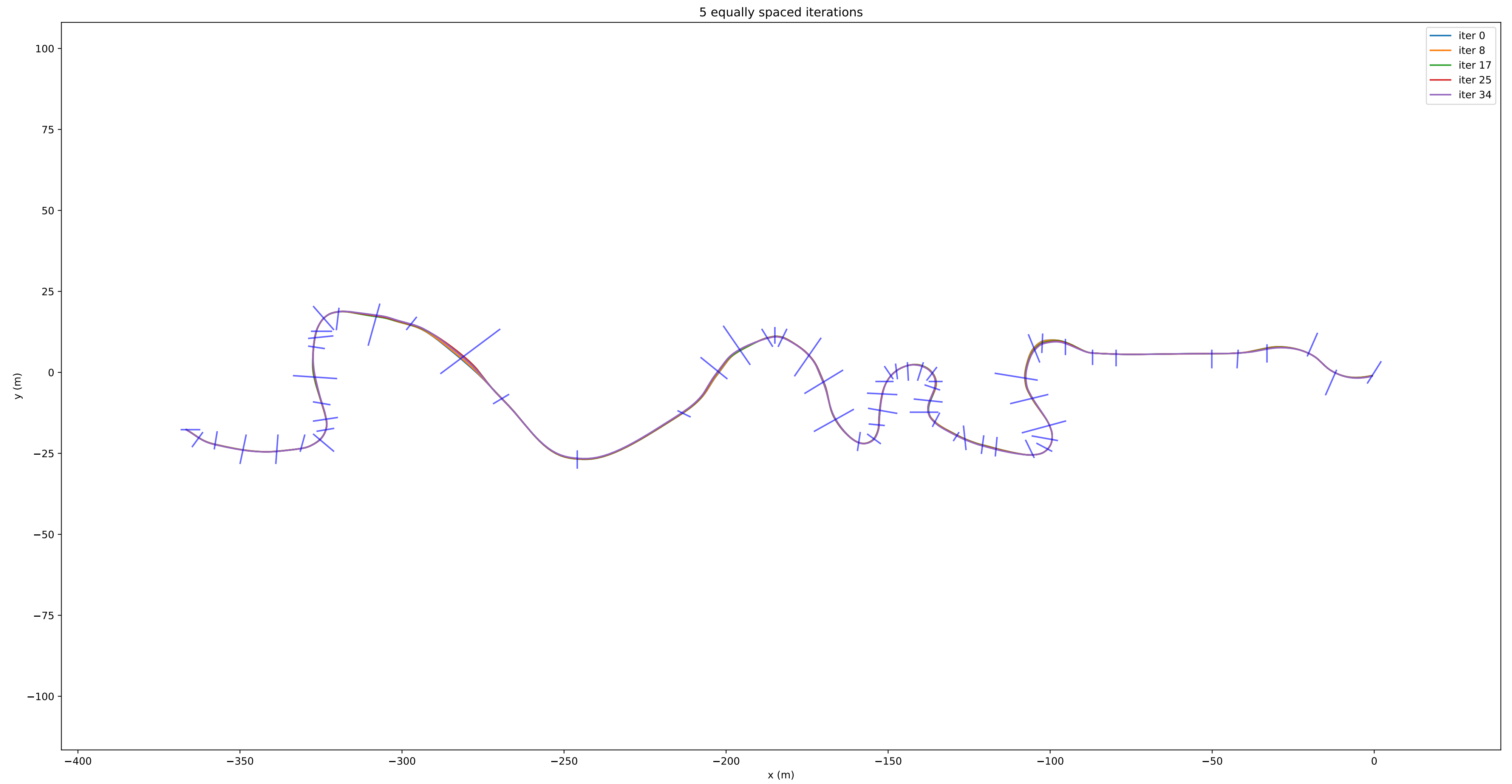
curvature objective function: 61.79 ms

Optimization progression

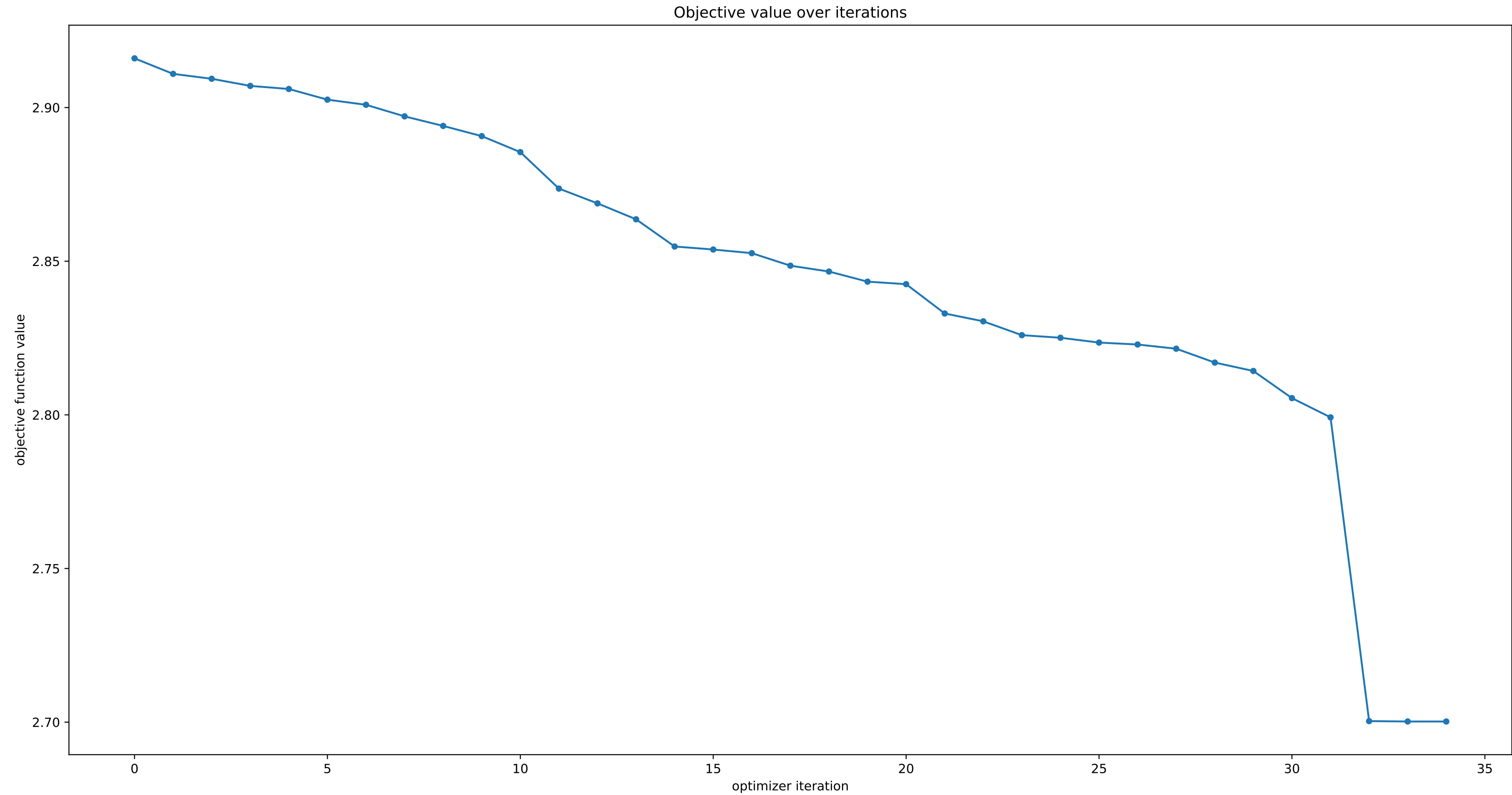
Optimization progression (35 iterations)



Equally spaced iterations



Objective value over iterations



Instant radius vs distance

